MERAKI - BIOS : RE-NEET 2026
TARGET 360/360 IN BIOLOGY
THREE [3] ALL INDIA FREE TESTS IN BIOLOGY
27.05.2026; 07.06.2026, 17.06.2026

PRACTICE TEST: REVISE AND RECALL: PHOTOSYNTHESIS IN HIGHER PLANTS

1. A first action spectrum of photosynthesis was described by:

1. Jan Ingenhousz
2. T.W. Engelmann
3. Julius von Sachs
4. Joseph Priestley

[PAGE 135 - Recall - Contribution of Cornelius van Niel]

2. There is a clear division of labour within the chloroplast where:

A: The membrane system is the site for enzymatic reactions that synthesise sugar, which in turn forms starch.
B: Stroma is responsible for trapping the light energy and also for the synthesis of ATP and NADPH.

1. Only A is correct
2. Only B is correct
3. Both A and B are correct
4. Both A and B are incorrect

[Page 136 - Recall - What are ‘dark reactions’ and why are they so called]

3. Thylakoid pigments like chlorophyll b, xanthophylls and carotenoids, which are called accessory pigments:

A: enable a wider range of wavelengths of incoming light to be utilised for photosynthesis.

B: protect chlorophyll a from photo-oxidation.

1. Only A is correct
2. Only B is correct
3. Both A and B are correct
4. Both A and B are incorrect

[Page 136 - Recall - What are ‘dark reactions’ and why are they so called]

4. Identify the correct statements:

A: Light reactions ['Photochemical' phase] include light absorption, water splitting, oxygen release, and the formation of high-energy chemical intermediates, ATP and NADPH.

B: The photosynthetic pigments are organised into two discrete photochemical light harvesting complexes (LHC) within the Photosystem I (PS I) and Photosystem II (PS II).

C: Each photosystem has only chlorophyll a forming a light harvesting system also called antennae.

D: In PS II the reaction centre chlorophyll a has an absorption peak at 700 nm, and is called P700.

1. Only A and B
2. Only A and C
3. Only B and D
4. Only C and D

[Page 138; Recall the basis of naming the two PS; structural framework of LHC]

5. Consider the two statements:

Assertion [A] : The protons and oxygen formed by the splitting of water during oxygenic photosynthesis is higher plants are released in the lumen of the thylakoid.

Reason [R]: The water splitting complex is associated with the PS II, which itself is physically located on the inner side of the membrane of the thylakoid.

1. Both A and R are true and R explains A
2. Both A and R are true, but R does not explain A
3. A is true but R is false
4. A is false but R is true

[Page 139; Recall - What replaces electrons that were moved from photosystem I?]

6. Identify the correctly matched pairs:

	Step of Calvin Cycle	Feature
A	Carboxylation	Catalysed by the enzyme RuBP carboxylase which results in the formation of two molecules of 3-PGA.
B	Reduction	Utilisation of two molecules of ATP for phosphorylation and one of NADPH for reduction per CO_2 molecule fixed
C	Regeneration	Requires two ATP for phosphorylation to form RuBP per CO_2 molecule fixed

1. Only A
2. Only A and B
3. Only B and C
4. A, B and C

[Page 144; Recall - how many ATP and NADPH molecules will be required to make one molecule of glucose through the Calvin pathway?]

7. The basic pathway that results in the formation of the sugars in photosynthesis, the Calvin pathway:

A: is common to the C_3 and C_4 plants.
B: occurs in all the mesophyll cells of the C_3 plants.
C: takes place only in the bundle sheath cells in C_4 plants.

1. Only A is correct
2. Only B and C are correct
3. A, B and C are correct
4. A, B and C are incorrect

[Page 147; Recall - Kranz Anatomy]

8. Consider the two statements:

Statement I: In C_4 plants photorespiration does not occur.
Statement II: In C_4 plants, RuBisCO functions as a carboxylase and lacks the oxygenase activity.

1. Statement I is correct; Statement II is correct
2. Statement I is correct; Statement II is incorrect
3. Statement I is incorrect; Statement II is correct
4. Statement I is incorrect; Statement II is incorrect

[Page 147; Recall - PEPCase; Complete Table 11.1 - Page 148]

9. Identify the incorrect statement:

1. Except for plants in shade or in dense forests, light is rarely a limiting factor in nature for photosynthesis in higher plants.
2. At low light conditions only C_3 plants respond to high CO_2 conditions.
3. Tropical plants have a higher temperature optimum for photosynthesis than the plants adapted to temperate climates.
4. The effect of water as a factor is more through its effect on the plant, rather than directly on photosynthesis.

[Page 150 - Recall - Blackman's (1905) Law of Limiting Factors]

10. [Challenger - Not 'lines of NCERT' but 'based and derived from NCERT']
Which of following statements regarding photosynthesis will be true?

A: In photosynthetic bacteria, the proteins that gather light for photosynthesis are embedded in cell membranes.
B: Chloroplasts may be photosynthetic bacteria that adapted to life inside plant cells.
C: In the reaction: $2\ H_2O + 2\ A + (\text{light, chloroplasts}) \rightarrow 2\ AH_2 + O_2$; A is the electron acceptor.
D: Phosphoglycolate produced during photorespiration cannot be metabolized by the Calvin-Benson cycle and represents carbon lost from the cycle.

1. Only A, B and C
2. Only A, C and D
3. Only B, C, and D
4. A, B, C and D

[Recall chromatophores, endosymbiosis; photolysis of water; products of oxygenase activity]